

Aylin García Soto

PERSONAL INFORMATION

Address: 2110 W University Dr, Boise, ID 83725, office - MP412

Phone: 857-318-2716

Email: aylingarciasoto@boisestate.edu

Website: aylingarciasoto.com

RESEARCH INTERESTS

Observations of M dwarf variability: I take spectroscopic data from the MDM Observatory telescopes and use photometric data from TESS to characterize the magnetic activity of M dwarf stars. For example, I compare contemporaneous Balmer emission lines ($H\alpha$, $H\beta$, $H\gamma$, $H\delta$) to trace flares and chromospheric heating with starspot photometric rotational modulation. These time series observations give us information on the strength and complexity of M dwarf magnetic fields, and how the resulting flares and chromospheric activity can shape the atmospheres and habitability of orbiting exoplanets.

Exoplanet Systems: I also use TESS photometry to study the orbital architectures and evolution of exoplanet systems around cool stars, extending prior obliquity work to M dwarfs (<4000 K) that were underrepresented in earlier Kepler-based samples. This work tests whether transiting planets around cool, active stars reflects spin-orbit misalignment as seen in K and G dwarfs or if the result is a selection effect due to high magnetic activity. I also observe and examine which exoplanet systems show signs of tidal decay as well as investigate the stellar parameters of their host star. In this project, I mentor undergraduate students in conducting observations, creating presentations and posters.

Citizen Science Project: I am leading a citizen science program that engages Idaho teachers in monitoring flares in young open clusters using a network of Unistellar eQuinox2 telescopes, funded by a NASA Science Activation grant. I coordinate our participating teachers, who use these scopes to gather flare data. The aim of the project is to strengthen our understanding of magnetic activity in young stars.

EDUCATION

Ph.D., Dartmouth College, Astrophysics <i>Advisor: Prof. Elisabeth Newton</i>	SPRING 2026
M.S., Dartmouth College, Astrophysics <i>Advisor: Prof. Elisabeth Newton</i>	SPRING 2023
B.A., Wesleyan University, Major: Astronomy, Minor: Planetary Science <i>Advisor: Prof. Edward Moran</i>	SPRING 2018

APPOINTMENTS

Postdoctoral Researcher - Boise State University	MARCH 2026 - PRESENT
Graduate Student Researcher - Dartmouth College	2019-2026
NASA ATP Executive Secretary - Virtual	2023
Graduate Student Council Representative/Leader	2020-2023
Post-Baccalaureate Researcher - Wesleyan University	2018-2020
TESS Objects of Interest (TOI) Deputy Manager - TESS MIT Office	2018-2019
McNair Undergraduate Researcher - Wesleyan University	2015-2018
Banneker & Aztlán Summer Researcher - Harvard CFA	2017
KNAC Summer Researcher - Williams College	2015

AWARDS

AAS Famous Grant	2026
Dartmouth College Conference Graduate Student Travel Award	2025
Matt Johns Astronomy Graduate Student Fund	2024
The Selamawit Tsehay Teaching Award	2021
NSF GRFP Honorable Mention	2021
E.E. Just Liftoff Fellow	2019-2026
AAS Famous Grant	2019
Wesleyan University Honors Thesis	2018
Wesleyan University Department of Astronomy's Littell Prize	2018
Ronald E. McNair Scholars Program	2015-2018
NASA CT Space Grant Scholarship	2015
Janey Scholars Program	2014-2018

TELESCOPE PROPOSALS & OBSERVING EXPERIENCE

Awarded Proposal:

TESS Guest Investigator Proposal (PI: William Waalkes, Co-I: Aylin García Soto) 2024

Awarded Observing Time:

MDM McGraw-Hill 1.3m Telescope (PI: Aylin García Soto) 29 Nights

MDM McGraw-Hill 2.4m Telescope (PI: Aylin García Soto) 32 Nights

Unawarded:

JWST Observing Proposal (PI: Benjamin Rackham, Co-I: Aylin García Soto) 2025

NSF NOIRLab Observing Proposal (PI: Sarah Logsdon, Co-I: Aylin García Soto) 2025

ANALYSIS EXPERIENCE

Data Analysis Experience: Optical photometry, Optical spectroscopy, X-ray photometry.

Software Experience: Python, IRAF, HTML

Software Contribution (Python packages):

- *protify* - TESS rotation periods
- *starrotate* - TESS rotation periods
- *TESS-ExoClass* - TESS Robo vetter.

NASA MISSION MEMBERSHIP

NASA SmallSat Pandora Science Team Member	2025 - PRESENT
NASA/MIT TESS Science Team Member	2019 - PRESENT

PUBLICATIONS

First-Author Publications

- **Aylin García Soto**, Melinda Soares-Furtado, Brian Jackson, et al. (**in prep**) “Is Wasp-12 a Subgiant?”
- **Aylin García Soto**, Elisabeth R. Newton, Benjamin V. Rackham, et al. (Aug. 2026). “Revisiting Photometric Variability as a Probe of Spin–Orbit Alignment with Kepler and TESS”. In: *Submitted to The Astrophysical Journal*.

- **Aylin García Soto**, Girish M. Duvvuri, et al. (Apr. 2025). “Short-term Balmer Line Emission Variability in M Dwarfs”. In: 982.2, 98, p. 98. doi: 10.3847/1538-4357/adb615.
- **Aylin García Soto**, Elisabeth R. Newton, Stephanie T. Douglas, et al. (May 2023). “Contemporaneous Observations of H α Luminosities and Photometric Amplitudes for M Dwarfs”. In: 165.5, 192, p. 192. doi: 10.3847/1538-3881/acc2ba.
- **Aylin García Soto**, Aleezah Ali, et al. (Apr. 2020). “Evidence for Transparency and Clumps in the Circumbinary Ring of the T Tauri Star V582 Mon (KH 15D)”. In: 159.4, 135, p. 135. Doi: 10.3847/1538-3881/ab6efd.

Collaboration Publications

- Sophia R. Flury et al. (**incl. Aylin García Soto**) (June 2026). “Activity and Structure of Low Mass Stars: Rossby Number, Mass, and a New Diagnostic”. In: Resubmitted to The Astrophysical Journal.
- Yoav Rotman et al. (**incl. Aylin García Soto**) (May 2026). “NASA’s Pandora SmallSat Mission: Simulated Modeling and Retrieval of Near-infrared Exoplanet Transmission Spectra”. In: 171.5, 263, p. 263. doi: 10.3847/1538-3881/ae4d3c. arXiv: 2603.04488 [astro-ph.EP].
- Keighley E. Rockcliffe et al. (**incl. Aylin García Soto**) (May 2026). “The unique ability of the Hubble Space Telescope to characterize young exoplanet environments”. In: arXiv e-prints, arXiv:2605.28768, arXiv:2605.28768. arXiv: 2605.28768 [astro-ph.IM].
- Girish M. Duvvuri et al. (**incl. Aylin García Soto**) (Oct. 2025). “FUMES IV: Optical and Far-ultraviolet Spectra of a Flare on the M Dwarf GJ 4334”. In: 170.4, p. 249. doi: 10.3847/1538-3881/adff74. arXiv: 2508.18459 [astro-ph.SR].
- Emily M. Boudreaux et al. (**incl. Aylin García Soto**) (Apr. 2024). “Correlations between Ca II H and K Emission and the Gaia M Dwarf Gap”. In: 965.1, p. 56. doi: 10.3847/1538-4357/ad2e8c. arXiv: 2402.14984 [astro-ph.SR].
- Michael Fausnaugh et al. (**incl. Aylin García Soto**) (Sept. 2021). “The TESS Mission Target Selection Procedure”. In: 133.1027, p. 095002. doi: 10.1088/1538-3873/ac1d3f. arXiv: 2109.02665 [astro-ph.IM].
- Natalia M. Guerrero et al. (**incl. Aylin García Soto**) (June 2021). “The TESS Objects of Interest Catalog from the TESS Prime Mission”. In: 254.2, p. 39. doi: 10.3847/1538-4365/abefe1. arXiv: 2103.12538 [astro-ph.EP].
- Pamela Rowden et al. (**incl. Aylin García Soto**) (Aug. 2020). “TIC 278956474: Two Close Binaries in One Young Quadruple System Identified by TESS”. In: 160.2, p. 76. doi: 10.3847/1538-3881/ab9d20. arXiv: 2006.08979 [astro-ph.SR].
- H. Parviainen et al. (**incl. Aylin García Soto**) (Jan. 2020). “MuSCAT2 multicolour validation of TESS candidates: an ultra-short-period substellar object around an M dwarf”. In: 633, A28. doi: 10.1051/0004-6361/201935958. arXiv: 1911.04366 [astro-ph.EP].
- Daniel Huber et al. (**incl. Aylin García Soto**) (June 2019). “A Hot Saturn Orbiting an Oscillating Late Subgiant Discovered by TESS”. In: 157.6, p. 245. doi: 10.3847/1538-3881/ab1488. arXiv: 1901.01643 [astro-ph.EP].

- Joseph E. Rodriguez et al. (**incl. Aylin García Soto**) (May 2019). “An Eccentric Massive Jupiter Orbiting a Subgiant on a 9.5-day Period Discovered in the Transiting Exoplanet Survey Satellite Full Frame Images”. In: 157.5, p. 191. doi: 10.3847/1538- 3881/ab11d9. arXiv: 1901.09950 [astro-ph.EP].
- Diana Dragomir et al. (**incl. Aylin García Soto**) (Apr. 2019). “TESS Delivers Its First Earth-sized Planet and a Warm Sub-Neptune”. In: 875.2, p. L7. doi: 10.3847/2041-8213/ab12ed. arXiv:1901.00051 [astro-ph.EP].

RESEARCH TALKS

Seminars & Colloquia

- 247th Meeting of the American Astronomical Society, Thesis Talk: “Short-term Magnetic Fields Activity on M-type Cool Stars And Its Effect on Exoplanet Detection” García Soto, A. et al. 2026.
- University of Arizona Origins Seminar (Link: https://youtu.be/qK_hc5UG8wY), “Magnetic Fields Activity on M-type Cool Stars Using TESS.” October 6, 2025. Host: Prof. Brittany Miles.
- Black Space Week: AAS Journal Author Series (Link: <https://youtu.be/uhpeE1V0vCM>), “AAS Journal Author Series: Aylin Garcia Soto on 2025ApJ...982...98G.” June 12, 2025. Host: Frank Timmes.
- Boston University Astronomy Colloquium, “Simultaneous Observations with TESS and OSMOS: Magnetic Fields Activity on M-type Cool Stars.” March 17, 2025. Host: Prof. Catherine Espilliat.
- University of Michigan APS Seminar, “Fully Convective M Dwarfs: Variability in Magnetically Sensitive Balmer Emission Lines.” García Soto, A., October 1, 2024. Host: Prof. Lia Corrales.
- University of Florida Astronomy Colloquium, “Simultaneous Observations with TESS and OSMOS: Magnetic Fields Activity on M-type Cool Stars.” August 29, 2024. Host: Prof. Sarah Ballard.
- University of Colorado Boulder APS Seminar, “Contemporaneous Observations of H α Luminosities and Photometric Amplitudes for M dwarfs.” June 23, 2023. Host: Prof. Zachory Berta-Thompson.
- TESS Mission Update Meeting (Science Team Meeting), “Contemporaneous Observations of H α Luminosities and Photometric Amplitudes for M Dwarfs.” 2023. MIT Building 6, Room 120.

Contributed Talks

- MIT Stars & Planets Workshop, “TESS Selection Effects and Stellar Obliquities in Fully Convective M dwarf Transit Hosts.”
- MIT Stellar Contamination Workshop, “Fully Convective M Dwarfs: Variability in Magnetically Sensitive Balmer Emission Lines.”
- TESS Science Conference 3 (TSC3), “Fully Convective M Dwarfs: Variability in Magnetically Sensitive Balmer Emission Lines.” (<https://zenodo.org/records/13136772>)

Posters

- 2026 Sagan Summer Workshop Exoplanets with Roman Surveys: Microlensing and Transits, “Citizen Science Project: Flares in Open Clusters.” García Soto, A. et al. 2026.
- Know Thy Star, Know Thy Planet 2 Conference, “M Dwarfs Variability in Magnetically Sensitive Balmer Emission Lines.” García Soto, A. et al. 2025.
- Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun. 22., “M Dwarfs Variability in Magnetically Sensitive Balmer Emission Lines.” García Soto, A. et al. 2024. (<https://zenodo.org/records/13118976>)
- Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun. 21., “Contemporaneous Observations of H α Luminosities and Photometric Amplitudes for M dwarfs.” García Soto, A. et al. 2022. (<https://zenodo.org/record/7523687>)

- 233rd Meeting of the American Astronomical Society, “Different Observational Properties of the Leading and Trailing Edges of the KH 15D Circumbinary Ring”, García Soto, A. et al. 2019.
- 231st Meeting of the American Astronomical Society, “Photometric Follow-up of Eclipsing Binary Candidates from KELT and Kepler”, García Soto, A. et al. 2018.
- 231st Meeting of the American Astronomical Society, “Uncovering extreme AGN variability in serendipitous X-ray source surveys”, E. C. Moran, García Soto, A. et al. 2018.
- APS Conference for Undergraduate Women in Physics at Harvard, “Dramatic X-ray Variability in AGNs: New Insights into Black Hole Fueling.” García Soto, A., and Moran, E. C. 2017.

ADVISING AND MENTORSHIP

SUNY Albany Science Research Program Mentor	JULY 2026 - PRESENT
The Short Period Planets Group Mentor	MARCH 2026 - PRESENT
Graduate Student Mentor for Dartmouth IGMP	2024-2026
Graduate Student Mentor for GSEC	2024-2026
Penpal for Letters for Pre-Scientist (LPS)	2023-2026
Newton Research Group: Chase Alvarado-Anderson	2020-2026
Dartmouth ASURE Graduate Student Panel	2022
Newton Research Group: Vera Krumm	2021-2022
Newton Research Group: Abigail Burrows	2020-2023
Graduate Student Mentor for E.E. Just Program	2020-2026
Women+ of Color Project Panels	2019-2020
Undergraduate Student Mentor for the McNair Program	2016-2018

SERVICE & OUTREACH EXPERIENCE

Public Lecture at Garden Valley Library	FALL 2026
Citizen Science Project - Flares Monitoring with Unistellar	MARCH 2026-PRESENT
Black In Space Week 2026	SUMMER 2026
IDAstro Outreach Mentor	MARCH 2026-PRESENT
Dartmouth Outreach Event Organizer	2019-2026
Dartmouth Telescope Operator and Trainer	2019-2026
Wikipedia Page KH 15D (V 582 Mon) Creator	2018
Wesleyan Outreach Organizer for Kids' Night and Space Night	2015-2017
Wesleyan Science Outreach Co-Coordinator	2015-2017

TEACHING APPOINTMENTS

Boise State University

Guest Lecturer - PHYS 305: Observational Astronomy (Background; SNR)	FALL 2026
Guest Lecturer - PHYS 405: Astrophysics (Stellar Spectral Types - Class 2)	FALL 2026
Guest Lecturer - PHYS 405: Astrophysics (Stellar Spectral Types - Class 1)	FALL 2026
Guest Lecturer - PHYS 405: Astrophysics (Parallax; Magnitudes - Class 2)	FALL 2026
Guest Lecturer - PHYS 405: Astrophysics (Parallax; Magnitudes - Class 1)	FALL 2026

Dartmouth College

Teaching Assistant - Astro 61: Observational Techniques in Astronomy	FALL 2025
Guest Lecturer - Astro 1 Exploration of the Solar System (Week 8/Day 3)	SUMMER 2025
Guest Lecturer - Astro 1 Exploration of the Solar System (Week 8/Day 1)	SUMMER 2025

Teaching Assistant - Astro 15: Stars and the Milky Way	SPRING 2025
Teaching Assistant - Astro 61: Observational Techniques in Astronomy	FALL 2023
Teaching Assistant - Astro 74: Astrophysics	FALL 2022
Teaching Assistant - Astro 2: Exploring the Universe	SUMMER 2021
Teaching Assistant - Astro 15: Stars and the Milky Way	SPRING 2021
Teaching Assistant - Astro 2/3: Exploring the Universe, with Laboratory	FALL 2020
Grader - Astro 2: Exploring the Universe	SUMMER 2020
Teaching Assistant - Astro 15: Stars and the Milky Way	SPRING 2020
Teaching Assistant - Phys 3: General Physics I	WINTER 2020

PROFESSIONAL REFERENCES

1. **Brian Jackson**

Professor of Physics and Astronomy
Boise State University
Email: bjackson@boisestate.edu

2. **Elisabeth Newton**

Professor of Physics and Astronomy
Dartmouth College
Email: Elisabeth.R.Newton@dartmouth.edu

3. **Zachory Berta-Thompson**

Professor of Astrophysics and Planetary Science
University of Colorado, Boulder
Email: zach.bertathompson@colorado.edu